Spatial Graph Attention and Curiosity-driven Policy for Antiviral Drug Discovery

ICLR
2022

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Paper

Code

Problem

Designing molecules that optimize a target property is hard: chemical space is discrete and vast, and molecular graphs resist faithful encoding.

> *Vast discrete space; molecular graphs resist faithful encoding.*

Motivation

- Prior models under-encode molecules, ignoring bond attributes and 3D structure, though shape decides binding.
- Atom-by-atom action spaces yield long, unstable trajectories and hard-to-synthesize molecules.

Contribution

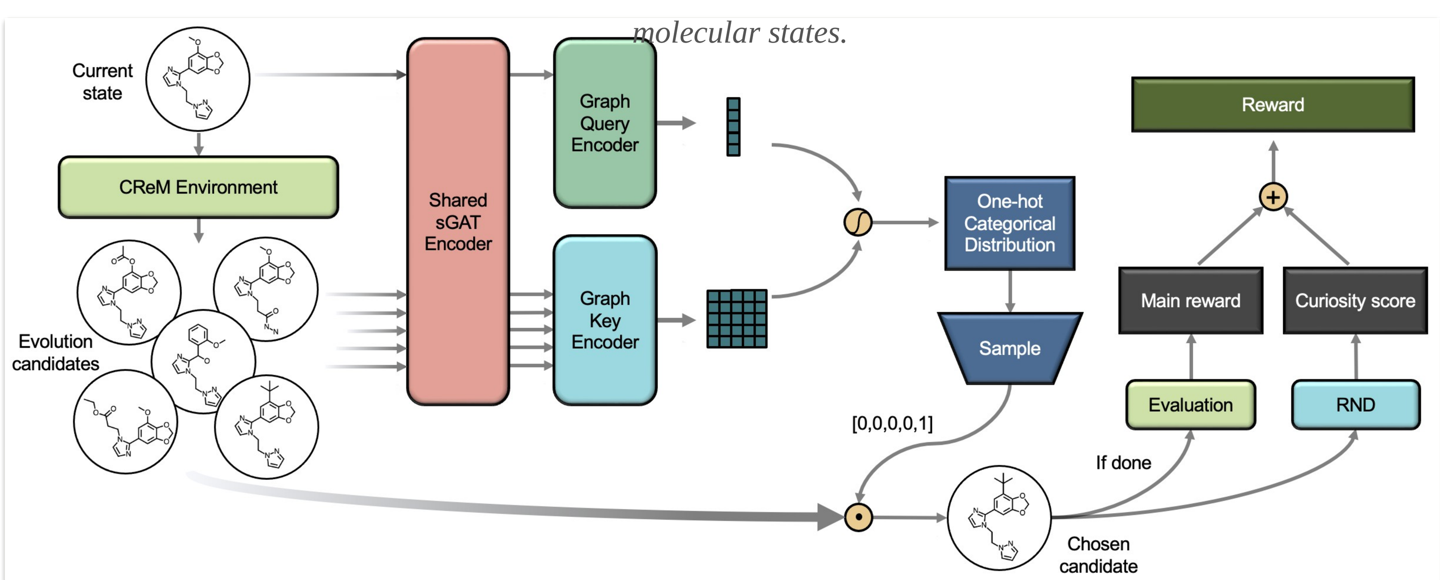
1. **sGAT** — self-attention over node & edge attributes plus 3D structure.
2. **GAPN** — attentional policy over a fragment-based action space (PPO).
3. **RND curiosity** — innovation rewards via random network distillation.

Method

- **Fragment MDP.** CReM proposes valid fragment-swap candidates; an attentional policy picks the next molecule — every step stays valid and synthesizable.
- **sGAT state encoder.** Self-attends over node/edge attributes and adds a spatial convolution on a sparsified inverse-distance matrix, fusing chemistry with geometry.
- **Curiosity-driven RL.** PPO actor-critic; random network distillation supplies an innovation reward driving exploration toward novel states.

$$r_i(s') = \mathrm{clip}_\eta\left(\|\hat{f}_\psi(s') - f^*(s')\| - m_b\right) / \sqrt{v_b}$$

Innovation reward: normalized RND prediction error rewards visiting novel molecular states.

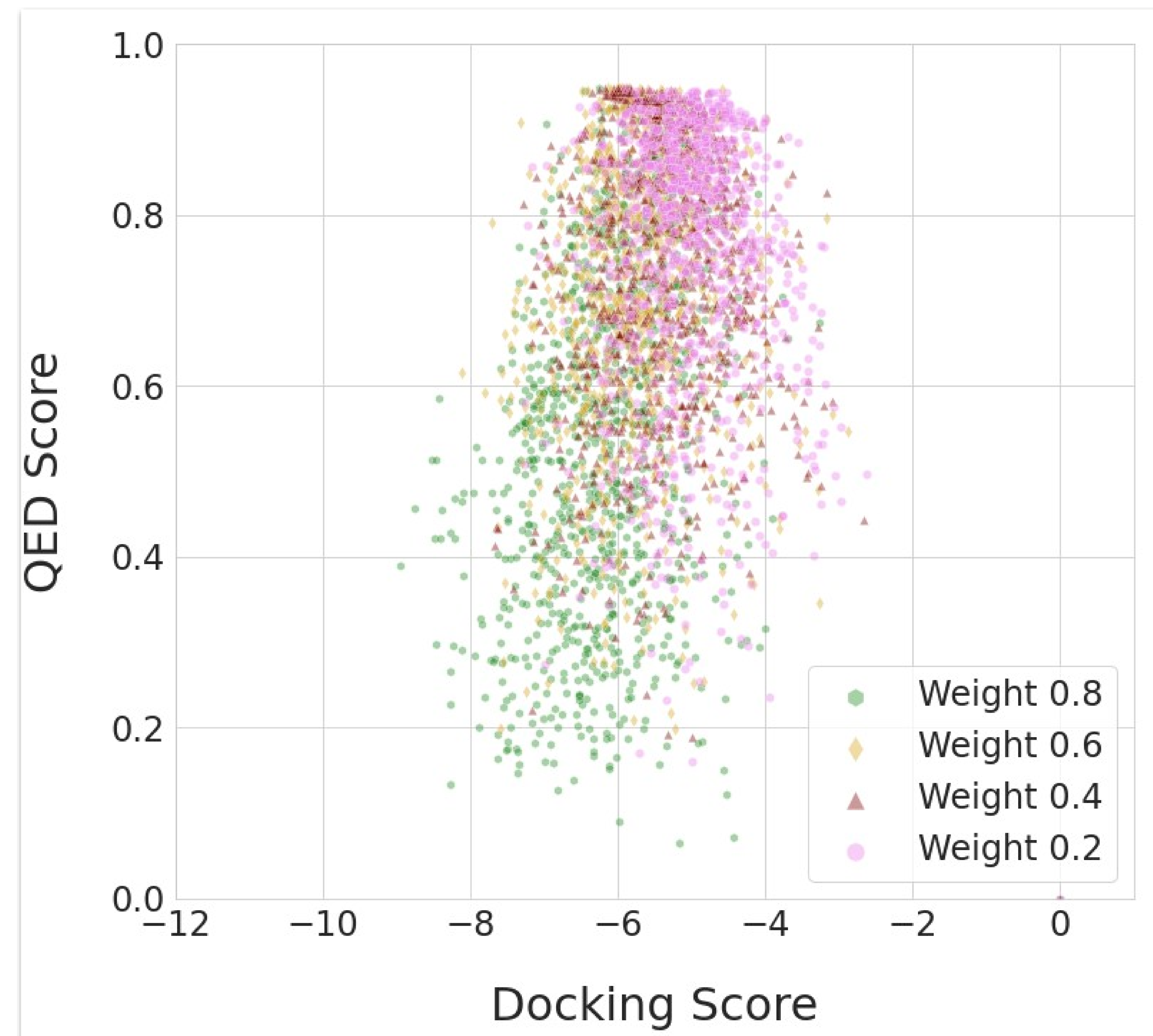


DGAPN in one generation step: CReM proposes candidates, a shared sGAT encoder feeds graph query/key encoders, the policy samples the next molecule, and reward combines evaluation with an RND curiosity score.

Key Results

NSP15 DOCKING	TOP ↓	MEAN ↓
MolDQN (2nd best)	−8.01	−5.38
DGAPN (ours)	**−10.07**	**−6.77**

−10.07 best docking vs MolDQN's −8.01 · p = 8.55×10^{-209} (Welch's t-test)



QED vs docking score for molecules under each weight ω — DGAPN keeps strong binders drug-like.

SO WHAT → *Beats five SOTA generators on binding affinity, with more synthesizable molecules.*

Ablation Study

- **Spatial channel.** Spatial convolution strongly improves supervised representation learning.
- **Curiosity.** Innovation rewards lift docking; DGAPN even beats a greedy docking oracle.

−10.07	−9.19	40
DGAPN top	no curiosity	sGAT runs

Headline Numbers

−10.07
Best docking score · NSP15
vs −8.01 for MolDQN (2nd best)

−6.77 mean docking

10^{-209} significance

0.72 QED (ω = 0.6)

Takeaway

By encoding 3D structure with spatial graph attention and rewarding novelty via RND curiosity, DGAPN generates higher-affinity, more synthesizable candidates than prior SOTA generators.

Spatial structure + curiosity → tighter binders and easier synthesis.